

Bluestock Fintech

Data Analyst Intern Project

Intern Details



Prepared By
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Host Organization
**Bluestock Fintech Pvt.
Ltd.**



Domain Focus **Mutual Fund /
Fintech Analytics**



Date of
Submission **June
2026**

Technologies and Tools Used :



Data Sources **AMFI India, mfapi.in,
NSE/BSE Public Portals**



CORE TECHNOLOGIES

Python

SQL

Pandas

Power BI

Tableau



Deliverables : **PDF Report + Dashboard
+ GitHub Repository**

Mutual Fund Analytics Platform

STRATEGY DEFINITION & ARCHITECTURE ROADMAP



Operational Bottlenecks

- ✗ **Scattered Data:** Key assets and metrics are isolated across fragmented AMC sheets, RTAs, and PDF reports.
- ✗ **Insight Latency:** Manual pipelines delay data consolidation, resulting in outdated portfolio metrics.
- ✗ **Risk Blindspots:** Absence of unified math layers prevents automated risk-adjusted comparisons.



Platform Objectives

- ✓ **Automated ETL:** Build robust pipelines for seamless data extraction, validation, and storage.
- ✓ **Single Source of Truth:** Unify underlying fund metadata and performance databases under one warehouse.
- ✓ **Interactive Intelligence:** Empower stakeholders with real-time portfolio health and risk dashboards.

Mutual Fund Analytics Platform

DATA SOURCES & PROCESSING PIPELINE



Source & Ingestion

Primary telemetry data was provided by **BLUESTOCK FINTECH**, representing historical and active performance cycles.

Downstream engineering operations performed on the raw files include structured **data cleaning**, schema type-casting, schema alignment, and statistical **quantitative analysis**.



Raw CSV Datasets

 `live_nav_hdfc_top100.csv`

 `live_nav_sbi_bluechip.csv`

 `live_nav_icici_bluechip.csv`

 `live_nav_axis_bluechip.csv`

 `live_nav_kotak_bluechip.csv`

 `live_nav_nippon_large_cap.csv`

System Architecture

A modular end-to-end pipeline designed for high-performance financial data processing, integrating disparate mutual fund datasets into a unified analytical engine.



1. Data Ingestion

- Multi-source CSV loading via Pandas
- Google Drive/Cloud integration
- Master-History relational mapping
- Handling AMFI code synchronization



2. Processing Engine

- Standardized cleaning protocols
- Forward-fill NAV missing values
- Datetime indexing & normalization
- Relational joins & aggregation



3. Analytics Layer

- Interactive Plotly time-series
- Market phase annotations
- Performance vs Benchmark alpha
- Demographic & Sector profiling

Market Trajectory & Demographics

EXPLORATORY DATA ANALYSIS: RETAIL GROWTH METRICS



Adoption Scale

Active industry folios nearly doubled from **13.26 Crore** to **26.12 Crore** over the evaluated period.

Systematic inflows peaked cleanly at a record terminal high of **₹31,002 Crore** in late 2025.



Youth Dominance

Millennial and Gen-Z buyers command the largest block, with investors aged **26–35** constituting **40.7%** of active profiles.

Median monthly ticket sizes remain highly uniform at **~₹5,000** across generations.



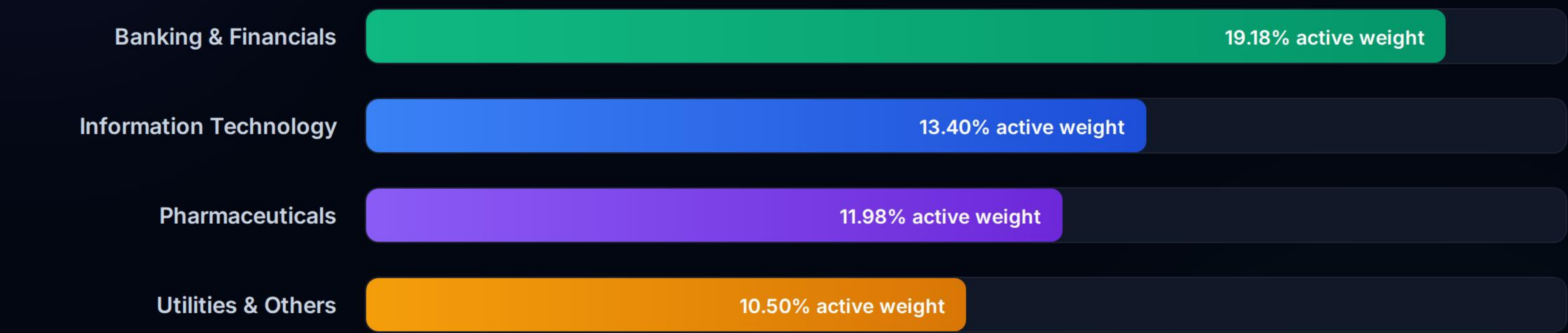
Gender Alignment

Investment registries expose a persistent structural balance gap, with male transactions capturing a **66.7%** share.

Active female investor ownership accounts for the trailing **33.3%** footprint.

Sector Allocations & Expense Drag

EXPLORATORY DATA ANALYSIS: PORTFOLIO WEIGHTS & COSTS



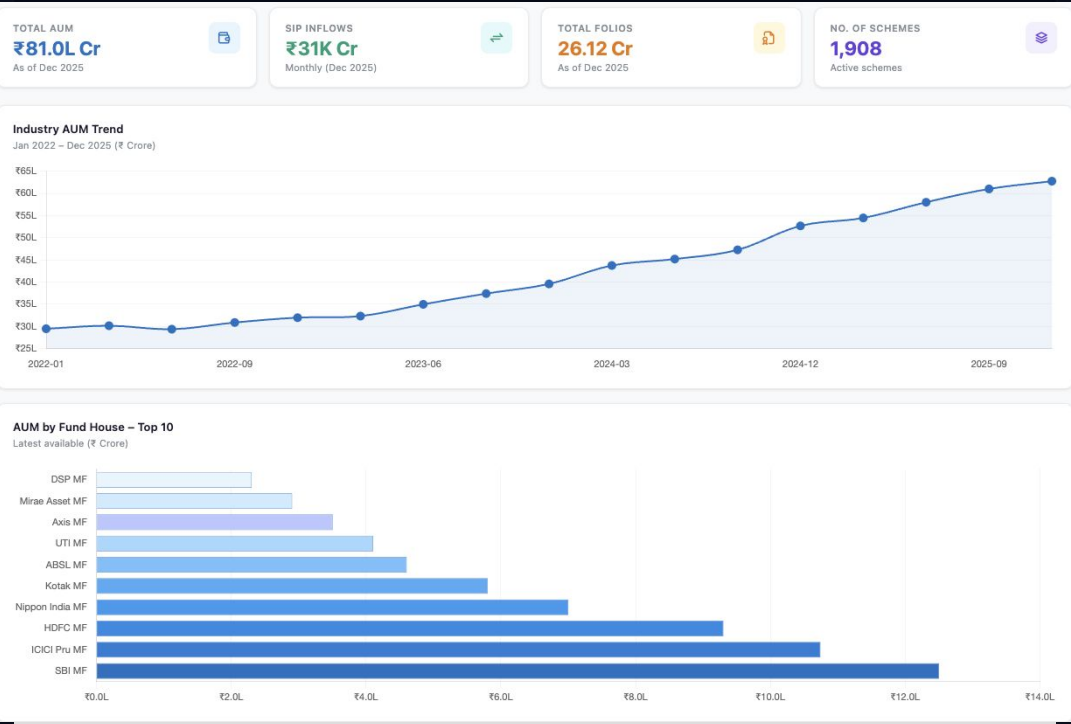
Key Analytics Takeaway: Consolidated equity portfolios reflect heavy structural reliance on financials and technology anchors. On cost distribution, regular plan retail setups expose trailing returns to systematic fee drags of **1.4% to 1.6%**, underperforming direct plans which optimize alpha generation.

Mutual Fund Portfolio Dashboard

PERFORMANCE MANAGEMENT SYSTEM: AUM SCALES & CORE SCORECARD METRIC MATRICES

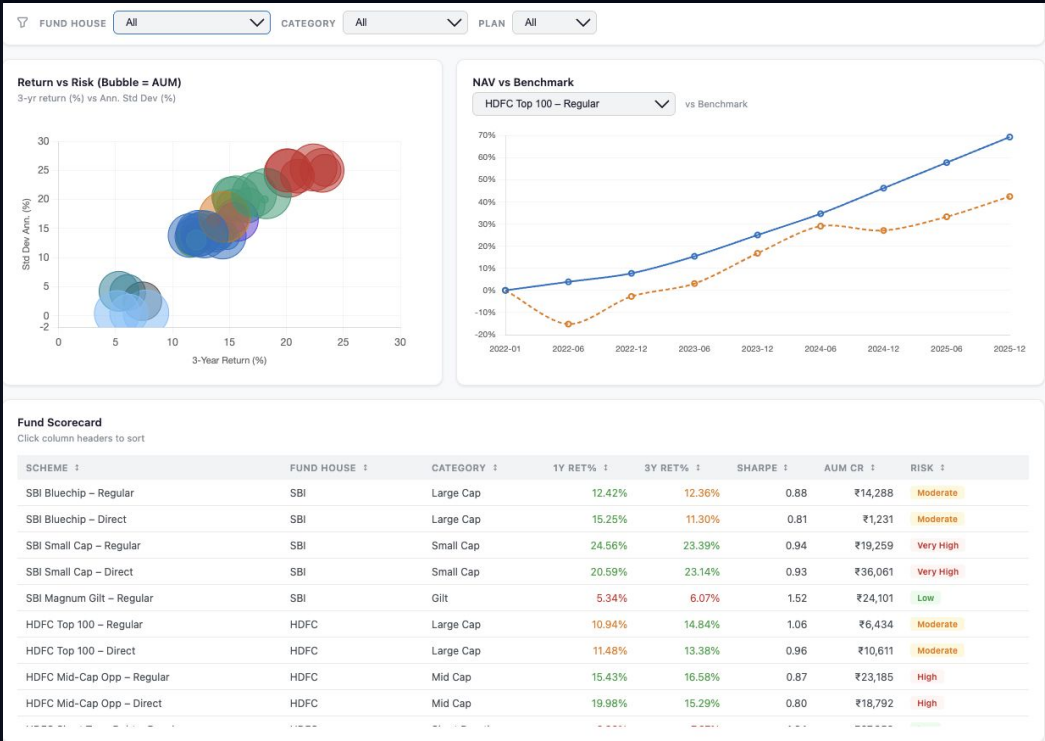
// Industry Inflows & AUM Rankings

dashboard 1_2.png



 Risk Return vs Benchmark Indices

dashboard 2_2.png





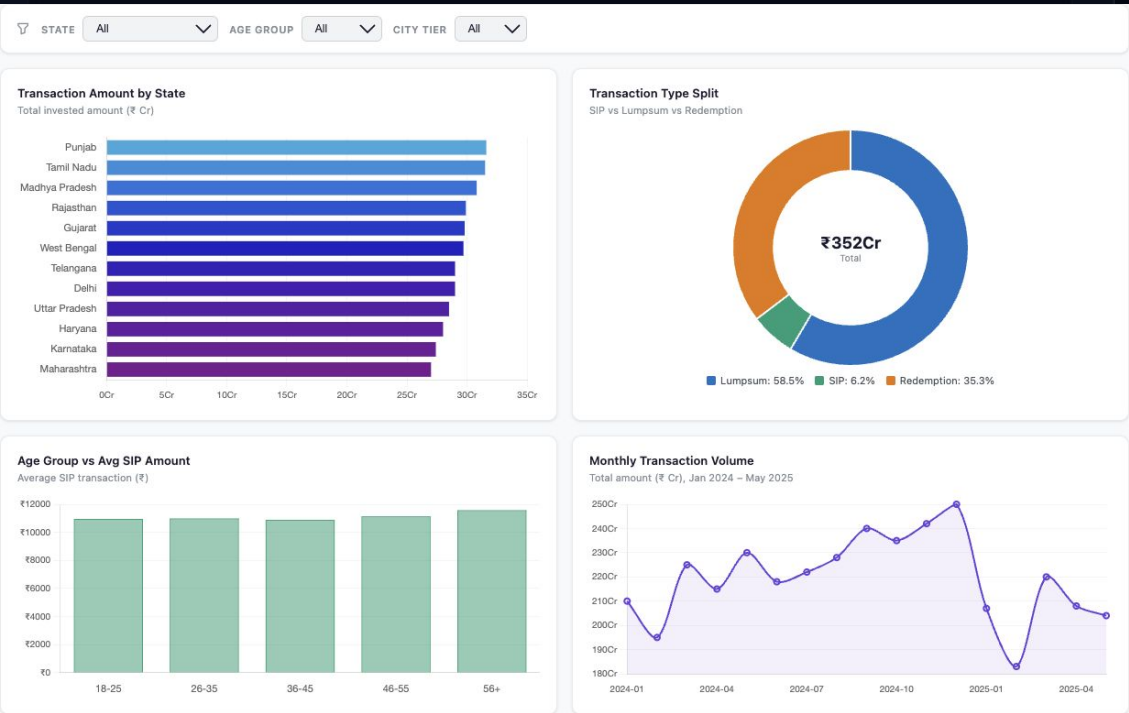
Investor Demographics & Inflows Dashboard

TRANSACTIONAL SEGMENTATIONS: REGIONAL DISTRIBUTIONS & SEASONAL HEATMAPS



Transaction Splitting & Demographic Triggers

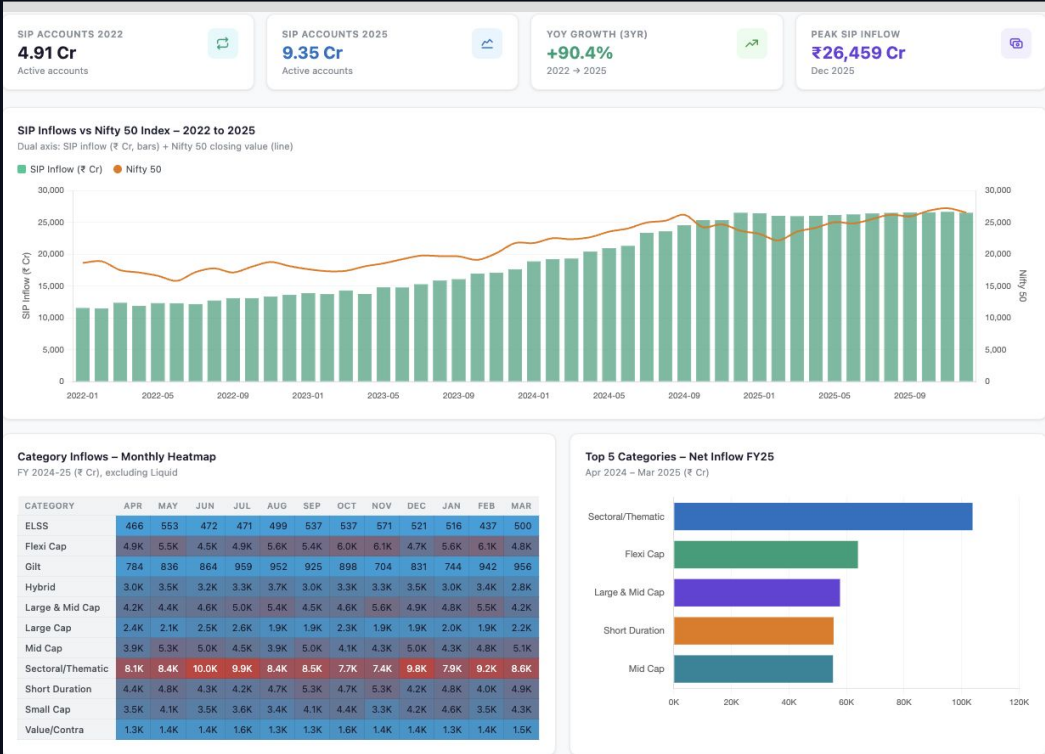
dashboard 3_2.png





Seasonality Maps & Monthly Heat Trends

dashboard 4_2.png



Executive Dashboard Insights

ANALYTICAL KEY FINDINGS & CORE PERFORMANCE METRICS



Exponential Growth Trajectory

Total Mutual Fund industry scale surged to ₹81.0L Cr by late 2025, supported by a massive +90.4% 3-year YoY growth in active retail SIP accounts (scaling from 4.91 Cr to 9.35 Cr).



Inflow Dominance & Assets

Lumpsum transactions command the highest transactional volume share at 58.5%. Strategic net inflows remain heavily concentrated in high-conviction Sectoral/Thematic and Flexi Cap schemes.



Balanced Retail Demographics

Capital allocation remains highly steady, showing a uniform average monthly ticket size of ~₹11,000 across all active age demographics, led regionally by Punjab and Tamil Nadu.



Thank You

